

1. Explain the specific consideration for the feasibility study report stage of a road project in hill regions in the comparison of terai (plane) one. [10]

Answer:

The feasibility study report stage of a road project in hill regions requires detailed investigation of topography, geology, slope stability, drainage and environmental impacts than a similar project in the Terai (plain) region because of the rugged and complex topography, unstable geological formations, higher altitude and so on.

The specific considerations for the feasibility study report stage of a road project in hill regions in comparison of terai region is shown below:

Hill regions	Terai region
i. As hilly regions consist of steep and rugged topography, irregular terrain, detailed study and contour mapping is essential.	i. As terai region consist of relatively flat terrain, the survey is comparatively simple.
ii. Alignment should follow contours as far as possible, where excessive gradients, sharp curves, landslide zones, unstable slopes need to be avoided.	ii. Alignment is generally straight and easy to selected with few constraints.
iii. Detailed geological investigation is required to identify weak rocks, faults, fractures, landslides and unstable slopes.	iii. Geological study in terai region mainly focus on the alluvial soil, bearing capacity, settlement and groundwater conditions.
iv. Stability of cuts and fill slopes, rockfall and landslide prone areas must be assessed.	iv. Local erosion and embankment instability may occur and needs to be assessed.
v. Drainage in hill road consists of side drains, catch-water drains, chute drains and cross-drainage structures.	v. Drainage mainly deals with floodwater, surface runoff and cross-drainage over relatively flat terrain.
vi. Numerous small streams, gullies and rivers may require culverts, causeways and bridges.	vi. The main concern in terai region is large river crossing and flood prone channels.

2. State importance of hill road drainage with examples. Show the surface drainage system for effective drainage of water with sketch. [10]

Answer:

Effective drainage is the most critical factor for the longevity and stability of hill roads. In the geologically fragile terrain of Nepal, water is the primary agent of destruction, making systematic drainage essential for protecting infrastructure and the environment. The importance of hill road drainage in Nepal are as follows:

- It prevents the road surface and subgrade from softening, which would otherwise lead to a loss of structural strength and bearing capacity.
- Proper drainage intercepts high-velocity surface runoff, preventing the formation of gully erosion and cross ruts on the pavement.
- By collecting and safely disposing of water, it protects embankments and road infrastructures from being damaged.
- In hilly areas, uncontrolled water increases pore water pressure within the soil mass, which is a leading trigger for landslides and slope failures.
- Subsurface drainage keeps the highest level of the groundwater table at least 1.2 metres below the subgrade, ensuring the stability of the foundation.

Some examples of hill road drainage in Nepal are as follows:

- i. **BP Highway (Sindhuli Road):** This is a commendable example in Nepal where a combination of cross-drainage, side slope drainage, and subsurface horizontal drainage is used to manage torrential monsoon rains and protect the mountainous terrain from water-induced landslides.



Fig: roadside retaining structure with the cascade drain for drainage in BP highway.

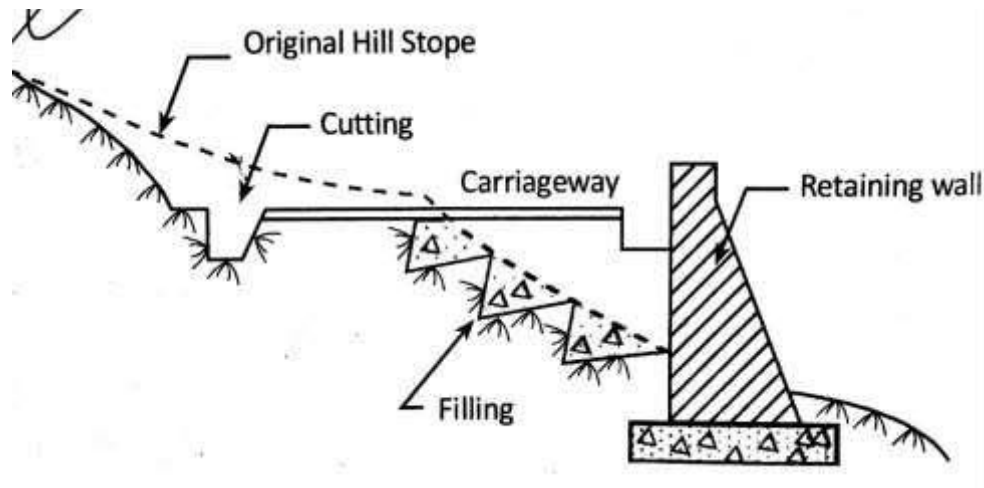
A hill road drainage system is designed to intercept and remove water from over, under, and the vicinity of the road surface to prevent the softening of the subgrade, loss of structural strength, and water-induced landslides. In mountainous regions like Nepal, effective drainage is critical for protecting the infrastructure from torrential monsoon rains and unstable geological conditions.

The hill road drainage system is categorized into two primary types: surface and subsurface drainage. The surface drainage system for the effective drainage is explained below.

Surface drainage components:

These components are responsible for removing and diverting water from the road surface and adjoining slopes:

- **Side Drains:** These are open channels provided along the edges of the road (typically on the hill side) to collect runoff from the pavement and cut slopes.
- **Catch-water Drains:** Located on the uphill side of the road, these drains intercept surface runoff from the mountain slopes before it reaches the roadway, diverting it into natural watercourses.
- **Chutes or Cascades:** Stepped structures used to safely convey collected water from higher-level catch drains down steep slopes to lower-level drains or outfalls.
- **Cross-Drainage (CD) Structures:** These allow water from natural streams to pass safely across the road alignment. They include:
 - **Culverts:** For waterways up to 6 m (pipe, box, slab, or arch culverts).
 - **Bridges:** For linear waterways greater than 6 m.
 - **Causeways:** Provided on less important roads where the depth of flow does not exceed 1.5 m.
 - **Aqueducts:** Conduits that drain water across the road from above, particularly useful in hilly areas where culverts are not feasible.



3. **How many types of cables are there in suspension bridge? Describe the function of each type. How is the load on the bridge transferred to the ground? Illustrate with simple sketches. [10]**

Answer:

A suspension bridge primarily uses three types of cables to support and stabilize the bridge. Each cable has a specific function in transferring loads safely to the towers and anchorages.

The main components of suspension bridge are given below:

i. Main cable (Suspension Cable):

- It is the primary load-carrying cable of the bridge.
- It passes over the tops of the towers and is anchored firmly at both ends.
- Usually made of thousands of high-strength steel wires bundled together.
- Carries the load of the bridge deck and traffic through the suspenders.
- Transfers the load to the towers and anchor blocks.

Function:

- Supports the entire bridge deck.
- Transfers loads to towers and anchorages through tension.

ii. Hanger cables (Suspenders):

- These are vertical steel cables or rods connecting the bridge deck to the main cable.
- Placed at regular intervals along the span.

Function:

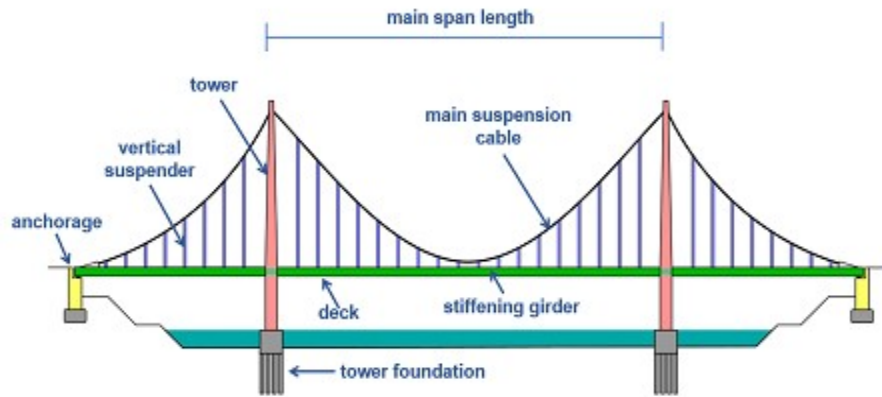
- Transfer the weight of the bridge deck and moving vehicles to the main cable.
- Keep the bridge deck suspended and properly aligned.

iii. Anchor cables:

- These are the portions of the main cable extending from the towers to the anchor blocks.
- They are firmly fixed into massive concrete anchorages.

Function:

- Balance the tension in the main cable.
- Transfer tensile forces safely to the ground through anchor blocks.
- Prevent the bridge from collapsing toward the center span.



In a suspension bridge, the load from vehicles and the bridge itself is transferred to the ground through a sequence of structural members.

Load Transfer Mechanism in flowchart diagram:

Traffic Load + Self-weight of Deck



Bridge Deck



Hanger Cables (Suspenders)



Main Cables



Towers (Pylons)



Anchor Blocks



Foundations



Ground (Soil/Rock)

- Bridge deck:
It carries traffic loads and transfers them to the hanger cables.
- Hanger cables:
Hanger cables also known as suspenders carry tension.
- Main cables:
Main cables also called suspension cables also carry tension.
- Towers:
They carry compression.
- Anchor blocks:
These blocks resist tension and prevent the main cables from pulling inward.
- Foundations:
Foundation of the bridge finally distributes all forces safely to the supporting soil or rock.

4. **What is meant by onsite sanitation system? Describe a septic tank for 200 users, rate of water supply 45 litres/capita/day, detention period 24 hours and cleaning of sludge as per 5 years. [10]**

Answer:

Onsite sanitation system refers to the collection, treatment, and disposal or reuse of wastewater and human excreta at or near the location where it is generated. In Nepal, approximately 40 out of every 100 people have access to safely managed on-site sanitation, and the sector is increasingly moving toward sustainable and resource-recovery models. Ecological sanitation (EcoSan), Septic tanks, greywater management are some of the common examples of onsite sanitation system.

Given:

Number of users, $P=200$

Water supply rate = 45 L/capita/day

Detention period = 24 hours = 1 day

Sludge cleaning interval = 5 years

Assuming 80% of the water supplied becomes sewage,

$$q=0.8 \times 45 = 36 \text{ L/capita/day}$$

Therefore,

$$Q=200 \times 36 = 7200 \text{ L/day} = 7.2 \text{ m}^3/\text{day}$$

For 24-hour detention:

$$V_d = Q \times 1 = 7.2 \text{ m}^3$$

Assuming sludge accumulation of 30 L/person/year:

$$V_s = 200 \times 30 \times 5$$

$$V_s = 30,000 \text{ L} = 30 \text{ m}^3$$

$$V = V_d + V_s$$

$$V = 7.2 + 30 = 37.2 \text{ m}^3$$

Assume:

Liquid depth = 1.5 m

Length: Breadth = 3: 1

Required plan area:

$$A = 1.5 / 37.2 = 24.8 \text{ m}^2$$

Let breadth = B , then length = $3B$.

$$3B^2 = 24.8$$

$$B = 2.88 \text{ m}$$

$$L = 3(2.88) = 8.64 \text{ m}$$

Hence, adopt approximately:

$$L = 8.6 \text{ m},$$

$$B = 2.9 \text{ m}$$

Provide 1.5 m liquid depth and about 0.3 m freeboard, giving a total internal depth of approximately: 1.8 m

5. **Explain the treatment process of waste in detail with sketches. [10]**

Answer:

The phenomenon in which impurities of wastewater are removed partially and changed to stable organic solids is known as wastewater treatment. It makes the effluent safe for the disposal and reduces its negative impact on surrounding environment and human health.

In sewage treatment process, there are three types of unit operations and processes. They are:

i. Physical unit operation:

- The operations used for the sewage treatment in which application of physical forces predominates is known as physical unit operation.
- It consists of:

- a. Screening:
 - It deals with removal of coarse and settleable solids by surface straining.
 - Different sizes of screens can be provided to trap and remove the floating debris and other settleable materials present in sewage.
- b. Skimming:
 - It deals with removal of oils and grease.
 - It consists of a rectangular tank in which air is blown to the sewage by an aerating device at the bottom, where the air bubbles coagulate, solidifies and causes it to rise on the surface.
- c. Grit chamber:
 - It deals with the removal of sand and grit.
 - The chamber separates the grit materials from organic solids by differential sedimentation.
- d. Primary sedimentation:
 - The process deals with removal of settleable solids and thickening of sludge.
 - It removes the fine suspended impurities heavier than water present in the sewage under the action of gravitational force.
 - Primary sedimentation removes 50%-70% of suspended solids from wastewater.

ii. Chemical unit process:

- The process used for sewage treatment in which the removal or conversion of contaminants is brought about by the addition of chemicals is known as chemical processes.
- It includes:
 - a. Chemical precipitation:
 - It is the method of removing suspended and colloidal solids in wastewater by the addition of coagulating chemicals.
 - Most common coagulants are aluminum sulphate, ferric salts, ferrous sulphates etc.
 - b. Gas transfer:
 - It includes addition and removal of gases.
 - c. Adsorption:
 - It includes removal of organics and used for de-chlorination of sewage before final discharge.
 - d. Disinfection:
 - It is the process of selective destruction of disease-causing micro-organisms.
 - e. De-chlorination:
 - It includes removal of chlorine residue that exists after chlorination.

iii. Biological unit processes:

- The process used for sewage treatment in which removal of contaminants is brought about by biological activity are known as biological unit processes.
- It consists of:
 - a. Aerobic process:
 - b. Anaerobic process:
 - c. Facultative process:

d. Anoxic denitrification:

- The process involves:
 - a. Conversion of dissolved and colloidal matter present in sewage to biological cell tissue and end products.
 - b. Removal of cell tissue usually by gravity settling in secondary settling tanks.

6. **Maintain principal components of hydroelectric schemes. Explain in detail about forebay, penstocks and intake structures with sketch. [10]**

Answer:

The principal components of hydroelectric schemes are as follows:

- i. **Diversion weir:**
It is the type of structure places across the river to divert a part of river flow tp the various water conveyance systems through intake.
- ii. **Intake:**
It is a structure to divert water into a conduit leading to the plant. Trash rack screen are provided to prevent entry of floating debris and coarse bed load into the water conveyance system. It consists of a control gate to regulate the flow through conveyance systems.
- iii. **Gravel trap:**
A gravel trap flushes out the bed sediments that enter into the approach canal back into the river. The gravel trap is constructed close to the intake in order to prevent gravel from getting into the approach channel. The main function of the gravel trap is to collect the bed load, smaller than the trash rack opening size entering through it to the approach canal.
- iv. **Settling basin:**
It is the structure to remove the suspended sediments from the conveyance water for the power plant. It shall be designed to ensure that the water entering the water conveyance system is free from sediments that can damage the penstock and turbine runners due to abrasion.
- v. **Headrace conveyance:**
It may be a pipe or tunnel or canal or combination of these for the purpose of delivering design flow to the forebay or surge tank with a minimum energy loss.
- vi. **Forebay:**
It is located at the beginning of the penstock shaft to supply the required flow to the turbine during start up, reducing the water hammer effect. It serves as a secondary settling basin as well and a sediment sluice to remove settled suspended particles.
- vii. **Surge tank:**
It is located between the headrace pressure conduit and steeply sloping penstock pipe. It also helps to eliminate water hammer effect and permit the liquid to surge in the tank. The main function of surge tank is to protect the long pressure tunnel and penstock pipe from high water hammer effect caused due to sudden acceptance and rejection of loads.
- viii. **Penstock:**
It is a pipe that conveys the flow from forebay or surge tank to the turbine and are designed to carry water to the turbines with least possible loss.
- ix. **Anchor blocks and supporting piers:**
It is an encasement of penstock designed to restrain the pipe movement in all durations. Anchor blocks should be placed at all sharp horizontal and vertical bends, since there are forces at such bends which will tend to move the pipe out of the alignment.

x. Powerhouse:

It is a structure complex where all the equipment for providing electricity is suitably arranged. Two basic requirements of powerhouse are functional efficiency and aesthetics.

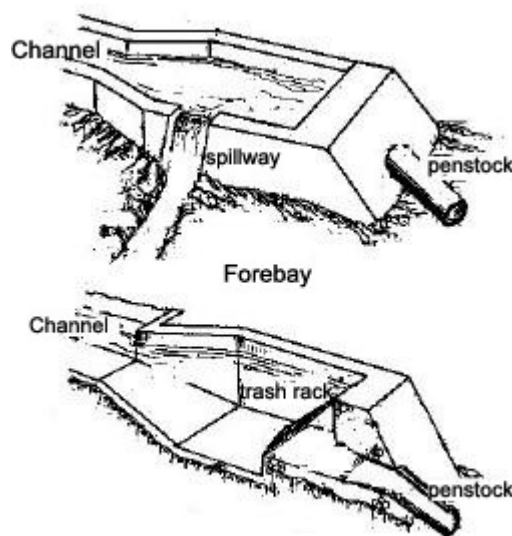
xi. Tailrace:

It disposes the design flow discharged by hydraulic turbine after power generation.

The explanation of forebay, penstocks and intake structures are given below:

i. Forebay:

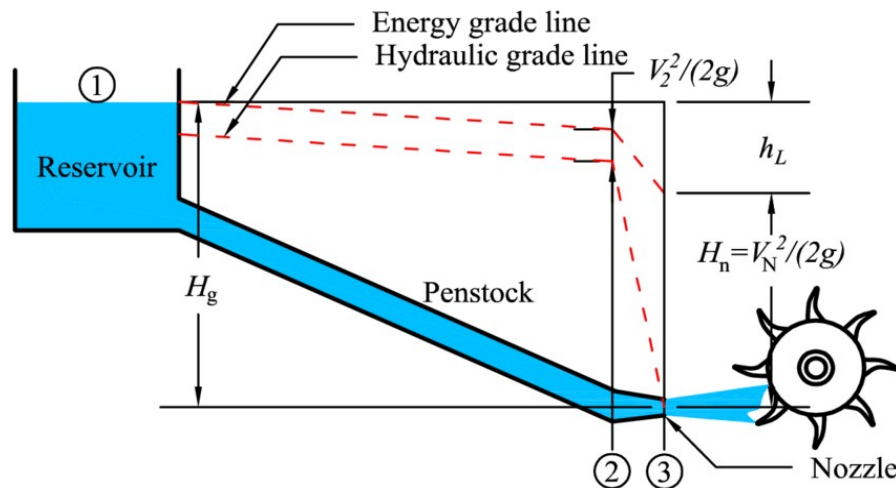
- It is located at the beginning of the penstock shaft to supply the required flow to the turbine during start up, reducing the water hammer effect.
- It serves as a secondary settling basin as well and a sediment sluice to remove settled suspended particles.
- Forebay is provided in the case of run-of-river plants and in case of storage plants when the power house is located certain distance away from the dam and water is carried from the reservoir to the power house through a channel.
- Forebay is similar to the settling basin except that the outlet transition is replaced by a trash rack.
- Functions of forebay:
 - a. To fulfill the immediate requirement of supplying water during turbine start-up.
 - b. To accommodate the rejected water during turbine closure and spill extra water.
 - c. Releases surge pressure as the wave travels out of the penstock pipe.
 - d. It also serves as a secondary settling basin and trap some particles that enter the headrace downstream of the settling basin.



ii. Penstocks:

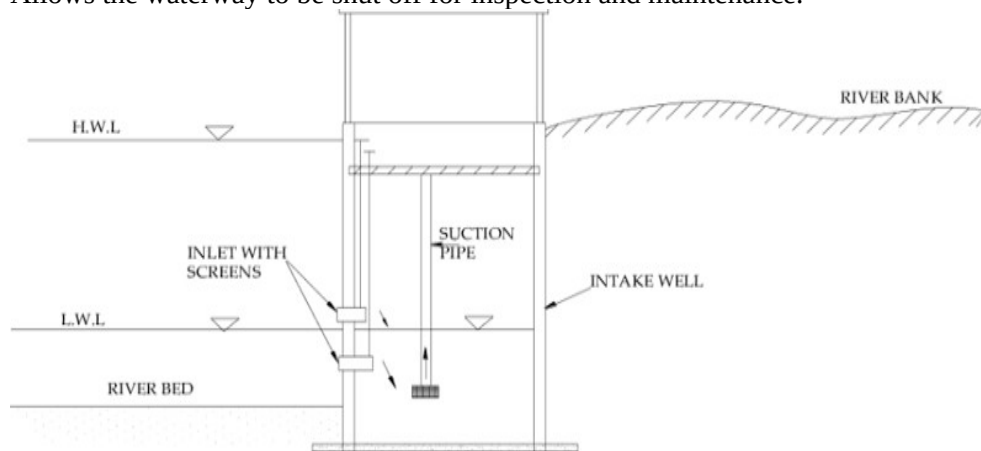
- These are the pipes of usually larger diameter usually of steel or concrete used for conveying water from reservoir or forebay or surge tanks to the turbine units.
- These are usually high-pressure pipelines designated to withstand static and water hammer pressures created by sudden changes in power demand.
- The penstock must withstand internal water pressure, water hammer, external loads and other structural stresses.
- Functions of penstock:
 - a. It conveys water from forebay to turbine of the hydropower plant.
 - b. It utilizes the available gross head for power generation.
 - c. It controls and directs water to the turbine.

- d. Penstock also provides a pressure-tight passage for high-velocity flow.



iii. Intake structures:

- An intake structure is the structure through which water is admitted from the reservoir, river or diversion pool into the water conveyance system of a hydropower project.
- It is one of the most important components because it must supply the required discharge while preventing debris, sediment and air from entering the conveyance system.
- Functions of intake structures:
 - a. Diverts the required quantity of water into the scheme.
 - b. Prevents entry of floating debris through trash racks.
 - c. Controls water flow using gates or valves.
 - d. Reduces entry of excessive sediment.
 - e. Allows the waterway to be shut off for inspection and maintenance.



7. **What do you mean by River Training works? Describe various types of river training works and protection works. [10]**

Answer:

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries.

Depending upon the purpose required river training works can be classified into three types.

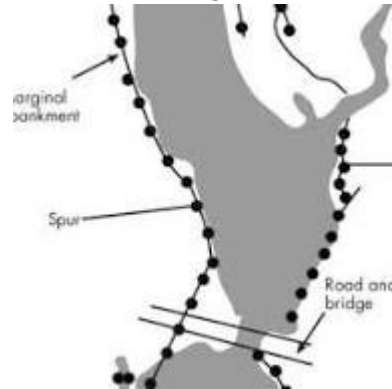
- i. high water training works: training for discharge

- ii. low water training works: training for depth
- iii. medium water training works: training for sediment

There are different methods of river training works. Some of them are described below:

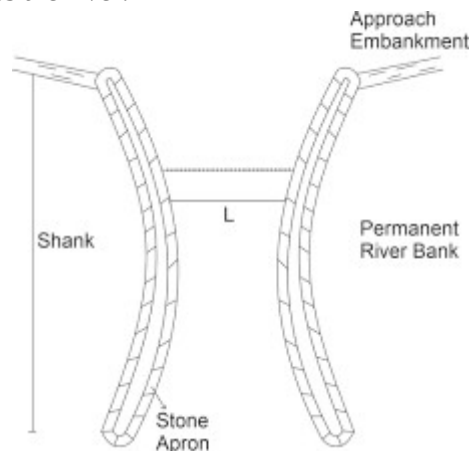
i. Marginal embankments/ levees:

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



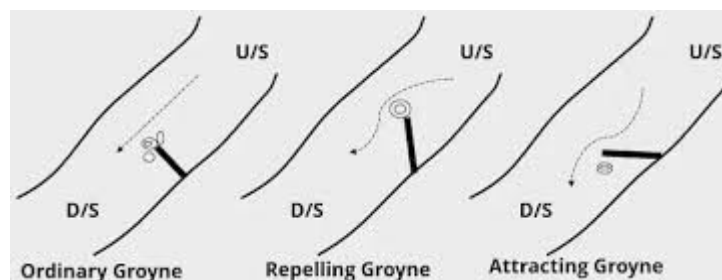
ii. Guide banks:

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



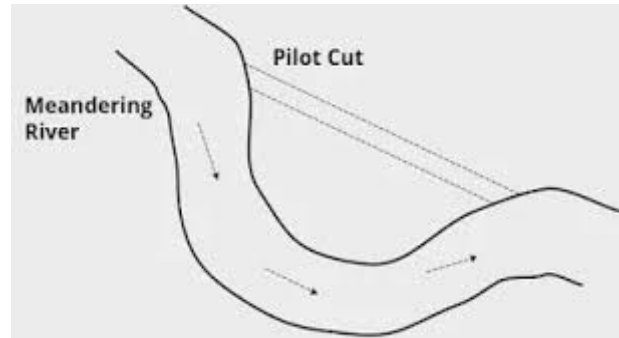
iii. Groynes or spurs:

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



iv. Artificial cutoff:

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.



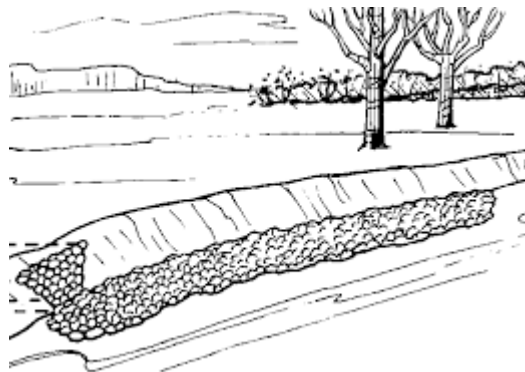
v. Bank protection and Pitched banks:

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

Bank protection work can be classified as direct or indirect.

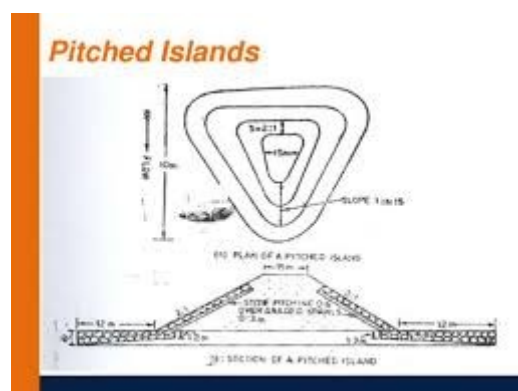
Direct protection works include work done on the bank itself such as vegetation cover, pitching

Indirect protection includes work constructed not directly on the banks but in front of them.



vi. Pitched islands:

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



vii. Bandalling:

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.



8. **Explain the specific considerations for planning, layout and design of the headworks in hill region as compared to those in the terai (plane) region. [10]**

Answer:

Headworks are the structures constructed at the head of a hydraulic system to divert, regulate and control water from a river into the system. In Nepal, the planning and design of headworks differ considerably between hill regions and Terai (plain) regions because of differences in topography, river characteristics, sediment load, floods and accessibility which is listed below:

Hilly region	Terai region
i. Hilly region has steep and irregular terrain so that headwork needs to be provided into narrow valleys and stable banks.	i. Terai region consists of flat and relatively uniform terrain which provides more space available for layout.
ii. River in hilly region has steep gradient and high flow velocity.	ii. River has relatively gentle slope and lower flow velocity.
iii. Coarse sediment, gravel, boulders and debris are common.	iii. Large quantities of fine sand, silt and suspended sediment are common.
iv. Canal must follow contour and is often located along the hill slopes.	iv. Canal can generally follow a relatively direct alignment over flat terrain.
v. Foundation may require rock anchoring, retaining structures and slope stabilization.	v. Foundation may be on deep alluvial deposits and seepage as well as piping are major concerns.
vi. Provision for the removal of coarse bed loads and boulders is important.	vi. Desilting arrangements for fine suspended sediment are important in terai region.
vii. Low accessibility to the site may increase construction and maintenance costs in hilly region.	vii. Better accessibility and easier transportation of construction materials in terai region.
viii. Construction work is difficult due to steep terrain, limited working space and difficult diversion of the river in hilly region.	viii. Construction is comparatively easier because of wider working areas, although large floods can complicate river diversion.

9. **“New technological Innovation has resulted a lot of benefit to the society”. Justify this statement with appropriate examples.**

Answer:

The development and application of new materials, methods, equipment, and systems to improve efficiency, quality, safety, and sustainability can be said as technological innovation. In this 21st century innovations and technological advancements are the crucial part of society. These innovations are developed to help humans do different works easily and rapidly. Technological advancements in the construction sector, have significantly improved people's quality of life and accelerated economic development.

The benefits of technological innovations are listed below:

- i. Improved efficiency in construction:
 - Modern equipment and machinery reduce the construction time and labor requirements.
 - Mechanized construction increases productivity and accuracy.
 - Example: Excavators, tower cranes, mix concrete plants etc.
- ii. Earthquake resistant construction:
 - New innovation and technologies also help to strengthen structural safety in earthquake-prone regions like Nepal.
 - Example: RCC framed structures, seismic resistant detailing, base isolation, energy dissipation device.
- iii. Better quality and durability:
 - Advanced construction materials improve the strength and lifespan of the structures.
 - Example: high-strength concrete, Fiber Reinforced Polymer, geotextiles etc.
- iv. Sustainable and environment-friendly development:
 - New technology also reduces the environmental impacts and promotes the efficient use of resources.
 - Examples: rainwater harvesting systems, green roofs, wastewater recycling, energy efficient buildings etc.
- v. Better water supply and irrigation:
 - It helps to improve the water conservation and agricultural productivity.
 - Examples: Drip irrigation, sprinkler irrigation, SCADA systems integration for water supply management.
- vi. Better project planning and management:
 - Computer based tools improve project scheduling, cost estimation and quality control.
 - Examples: AutoCAD, Civil 3D, Primavera P6, Microsoft Project etc.
- vii. Modern survey and mapping:
 - Survey work has become more accurate and less time-consuming.
 - Examples: Total Station, GNSS surveying, LiDAR technology.
- viii. Economic growth and employment:
 - Technology promotes industrialization, creates skilled jobs and increase the national productivity.
 - It supports infrastructure development and attracts investment.

10. Describe the different construction technologies that can be used in the construction of low-cost housing for urban poor. Mention indigenous technology to be adopted in building design and construction. [10]

Answer:

Low-cost housing provides safe, durable and affordable houses by reducing construction cost without compromising essential requirements of building. The following construction technologies can be used for construction of low-cost housing for the urban poor:

- i. **Use of locally available materials:**
Locally available materials such as stone, brick, bamboo, timber, earth and other materials to reduce transportation and material costs.
- ii. **Stabilized soil blocks:**
Compressed or stabilized earth blocks made by mixing soil with a small quantity of cement or lime can be used for walls.
- iii. **Rat-trap bond brick masonry**

Bricks are placed on the edge to create cavities within the wall, reducing brick and mortar consumption while providing good thermal insulation.

iv. **Bamboo and timber technology**

Properly treated bamboo and timber can be used for roofs, partitions, doors, windows and lightweight structures.

v. **Improved roofing technology**

Lightweight and economical roofing materials such as CGI sheets, fibre-cement sheets, tiles and locally available roofing materials can be used.

vi. **Water and sanitation technologies**

Rainwater harvesting, low-water-use sanitary fixtures, simplified drainage and decentralized sanitation can reduce both construction and operating costs.

Indigenous technologies are traditional, locally developed construction practices suited to local materials, climate and culture. The indigenous technologies to be adopted in building design and construction are mentioned as follows:

a. Timber bands:

- Horizontal timber bands are provided at plinth level, lintel level and roof level.
- These bands tie the walls together and distribute the earthquake forces uniformly.
- They reduce the chances of wall separation and collapse.



b. Timber frames with masonry infill:

- A wooden frame carries most of the structural load.
- Brick or stone masonry is used as infill.
- It provides flexibility and ductility.



c. Wooden pegs and traditional joints:

- Timber members are connected using wooden pegs and interlocking joints.
- The connections remain flexible and absorb the earthquake energy.
- It is less brittle than rigid connections.



d. Lightweight roof construction:

- Traditional roofs use timber trusses with slate or light roofing materials.
- Uniform load distribution minimizes torsional effects.
- Simple geometry performs the better during earthquakes.



- e. Thick stone and brick masonry walls:
 - Thick walls provide the stability against gravity loads.
 - It is combined with timber bands, which performs better than unreinforced masonry alone.

